



CODECHEF

Welcome to the CodeChef.com Challenge—your monthly dose of coding puzzles from India's biggest coding contest, now in print!

A whole month's gone by and once again it's time to compete in the CodeChef Puzzle Challenge. This is your chance to test your skills amongst other passionate programmers, and if you're right and lucky then you will be awarded by *LINUX For You* and Directi.

Last month's puzzle

How many numbers from 1 to 10,000 (inclusive) are divisible by 3 or 5 or 7?

The solution

There are 5,429 numbers from 1 to 10,000 (inclusive) that are divisible by 3 or 5 or 7.

How did we do it?

The simplest solution to this problem would be to write a program that marks off all the multiples of 3, 5 and 7. So we declare an array of appropriate size and initialise all elements to 0. Then, we make the elements at position 3, 6, 9... and so on as 1. We do the same for elements at positions 5, 10, 15... and similarly for multiples of 7. Now, we simply count the number of 1's in the

final array and we have our answer.

Alternatively, we can solve this problem using the inclusion-exclusion principle. Assume $F(a)$ is the number of integers less than 10,000 divisible by a . Then, by the inclusion-exclusion principle, the answer for the given question becomes $F(3) + F(5) + F(7) - F(3*5) - F(3*7) - F(5*7) + F(3*5*7)$. We can get the value of $F(a)$ by dividing 10,000 by a and taking the quotient. So, $F(3) = 10000/3 = 3333$ and so on.

And the December winners are...

- Manu Konchady
- Pawan Singh
- Jagannath G

Puzzle for January

A proper divisor of a number is a divisor that is less than the number itself. Find the sum of all proper divisors for the number 1,23,748.

For example, the proper divisors for the number 10 are 1, 2 and 5. So the answer would be $1 + 2 + 5 = 8$.

Need some extra cash?

Then solve this month's CodeChef Challenge puzzle and mail your answer to codechef@elyindia.com. Three lucky winners will win cash of Rs.1,000 each! 

About CodeChef: CodeChef.com is India's first, non-commercial, online programming competition, featuring monthly contests in more than 35 different programming languages. CodeChef has been created by Directi, as a way to continuously challenge and engage the developer community. The goal is to provide a platform for practice, competition and improvement as well as enable developers to benchmark their skills against their peers. Log on to www.codechef.com for the December Challenge that takes place from January 1-11, 2010, to win cash prizes up to Rs 55,000!